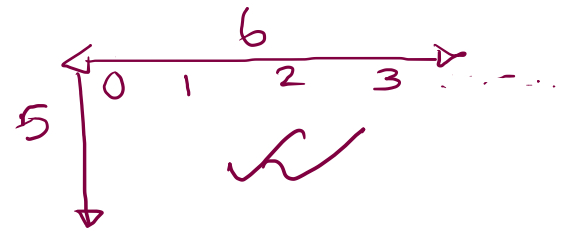


4.6 - Cryptography

* Classical Cryptography



i Caesar's Cypher

0 25
A → Z $(27+3) \bmod 26$

$p=3 \rightarrow 27$

$$f(p) = (p+3) \bmod 26 \quad 0 \leq x < d \rightarrow [0, d[$$

$$(p+3) \bmod (26 + \text{ASCII val})$$

ex 'MEET YOU'

$f(p)$ $\begin{matrix} 12 & 4 & 4 & 19 & 24 & 14 & 20 \\ 15 & 7 & 7 & 22 & 1 & 17 & 23 \end{matrix}$
"P H H W B R X"

$$J = 10^{\text{th}} \rightarrow (10-1) = [9]$$

$$M = 13^{\text{th}} \rightarrow (13-1) = [12]$$

$$P = 16^{\text{th}} \rightarrow [15]$$

$$F = 6^{\text{th}}$$

$$H = 8^{\text{th}} \rightarrow 7$$

"P H H W B R X"

$$f^{-1}(p) = (p-3) \bmod 26$$

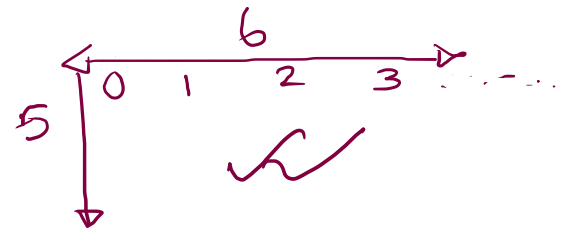
12 4 4 19 24 14 20

"MEET YOU"

if $K \neq 3 \rightarrow$ public-key Shifting.

4.6 - Cryptography

* Classical Cryptography



i Caesar's Cypher

0 25
A → Z (27+3) mod 26

p = 3 → 27

$$f(p) = (p+3) \bmod 26 \quad 0 \leq x < d \rightarrow [0, d[$$

$$(p+3) \bmod (26 + \text{ASCII val})$$

ex 'MEET YOU'

12 4 4 19 24 14 20
15 7 7 22 1 17 23
"P H H W B R X"

$$J = 10^{\text{th}} \rightarrow (10-1) = [9]$$

$$M = 13^{\text{th}} \rightarrow (13-1) = [12]$$

$$P = 16^{\text{th}} \rightarrow [15]$$

$$F = 6^{\text{th}}$$

$$H = 8^{\text{th}} \rightarrow 7$$

"P H H W B R X"
[f^-1(p)] = (p-3) mod 26
12 4 4 19 24 14 20
'MEET YOU'

if $K \neq 3 \rightarrow$ public-key Shifting.

② BLOCK CYPHER / Transposition

- Divide your msg into blocks of size $|m|$
- If # of letters in the msg. is not divisible by $|m|$ we add random letters at the end to make it divisible by $|m| \rightarrow |\sigma|$
- A transposition function $|\sigma|$ is defined for alphabet shifting.
- msg: PIRATE ATTACKS

$|msg| = 12$
 blocks of size (4) $m \rightarrow 4$
 $|\sigma|$

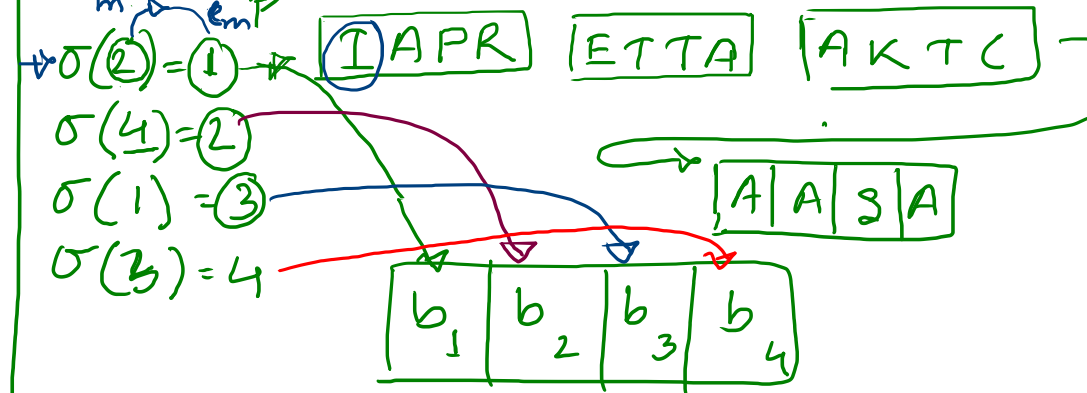
$\rightarrow \sigma = \{1, 2, 3, 4\}$
 $\sigma(1) = 3$ $\sigma(3) = 4$
 $\sigma(2) = 1$ $\sigma(4) = 2$

$4 \nmid 12$
 $|msg| = 13$
 $4 \nmid 13$ $4 \nmid 16$


PIRA TEAT TACK SAAA

(a)

(b) $\sigma(i) = x \in \sigma$
 $\sigma(2) = 1$ $\sigma(4) = 2$ $\sigma(1) = 3$ $\sigma(3) = 4$



(c) Decode

$$\begin{cases} \sigma(2) = 1 \\ \sigma(4) = 2 \\ \sigma(1) = 3 \\ \sigma(3) = 4 \end{cases}$$

→ TAPRETTA AKTCLASA
↓
PIRA|TEAT|TACK|SAAA

WPA2/AES

$$\begin{cases} \sigma^{-1}(1) = 2 \\ \sigma^{-1}(2) = 4 \\ \sigma^{-1}(3) = 1 \\ \sigma^{-1}(4) = 3 \end{cases}$$

$$\sigma\left(\underset{\uparrow}{i}\right) = \underline{m'}$$

$i \rightarrow$ index of letter in msg (E/D)
 $m' \rightarrow$ " " " " " (D/E)

